VCO2IL: Visual Computing (IM.ma VZ SS26)
Lab Report 1

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Abstract

ToDO: Write a brief abstract (max. 150 words) summarizing the main goals, methods and results of this assignment.

1.1 Related Work: Object Detection in Aerial and Satellite Imagery

1.1.1 Overview

Object detection in aerial and satellite imagery refers to the automatic localization and classification of objects of interest in overhead images of the Earth's surface. Typical target objects include vehicles, ships, airplanes, buildings, storage tanks, roads, and other man-made structures. This task is an important component of remote sensing image interpretation and supports a wide range of applications, such as urban planning, geographic information system updating, disaster response, precision agriculture, environmental monitoring, and security-related analysis [2, 4].

Although object detection has been widely studied in natural images, aerial and satellite imagery introduces several domain-specific characteristics that make the task fundamentally different from ground-level detection. Natural images usually capture objects from a horizontal perspective and show their profile information, whereas remote sensing images are acquired from an overhead perspective and mainly capture roof or top-view information [4]. As a result, object appearance, scale, orientation, and surrounding context differ substantially from standard computer vision datasets. In addition, aerial images often contain objects with arbitrary orientations, large scale variations, dense object distributions, complex backgrounds, and very large image sizes [2].

These properties make direct transfer of standard object detectors to aerial and satellite imagery difficult. For example, small vehicles may occupy only a few pixels, while airports or harbors can span large regions of an image. Objects such as ships, airplanes, and vehicles are often not aligned with the image axes, which motivates the use of oriented bounding boxes instead of conventional horizontal bounding boxes [2]. Furthermore, large satellite or aerial images often need to be processed using tiling or sliding-window strategies because they cannot be passed through detection networks at full resolution.

This chapter reviews the main challenges and methods for object detection in aerial and satellite imagery. First, it discusses important challenges such as scale variation, arbitrary object orientation, dense object distributions, large image sizes, and foreground-background imbalance. Second, it introduces oriented object detection and explains why horizontal bounding boxes are often insufficient for overhead imagery. Finally, relevant benchmark datasets and evaluation metrics are summarized, with particular attention to widely used datasets such as DOTA, DIOR, and xView.

1.1.2 Challenges in Aerial Imagery

A major challenge in aerial and satellite imagery is the strong variation in object scale. Small objects such as cars or small vehicles may occupy only a few pixels, while large structures such as airports, harbors, bridges, or industrial buildings can cover large image regions. This makes single-scale detection insufficient, because a feature representation that is suitable for large objects often lacks the spatial detail needed for small objects. Feature Pyramid Networks (FPN) address this problem by constructing multi-scale feature representations inside the network, allowing detections to be made at different feature levels [2, 3, 5].

Another important difference compared to ground-level imagery is the arbitrary orientation of objects. In aerial images, objects such as ships, airplanes, vehicles, or storage tanks are observed from a top-down perspective and are therefore not necessarily aligned with the horizontal or vertical image axes. Conventional horizontal bounding boxes (HBBs) can include large amounts of background for rotated objects and may localize them only imprecisely. For this reason, oriented bounding boxes (OBBs) are often more appropriate in aerial object detection, since they can represent the actual object extent more accurately [1, 2].

Dense object distributions further complicate detection. In scenes such as parking lots, harbors, airports, or urban traffic areas, many objects can appear close to each other or even overlap in the image. In such cases, horizontal boxes may cover multiple neighboring instances, which can lead to ambiguous feature extraction and localization errors. Moreover, post-processing methods such as non-maximum suppression become more difficult in high-density scenes, because correct detections of nearby objects may have high overlap and can be suppressed incorrectly [1, 2].

Large image sizes are another practical issue in aerial and satellite imagery. Remote sensing images can be much larger than typical natural images, and in Earth observation settings image sizes may reach tens of thousands of pixels in width and height. Such images cannot usually be processed directly by standard convolutional object detectors because of memory and computation limits. Therefore, large images are commonly divided into smaller tiles or processed using sliding-window strategies. However, this introduces trade-offs: overlapping tiles reduce the risk of cutting objects at tile borders but increase computation time, while smaller tiles may lose important contextual information [2].

Finally, aerial imagery often suffers from a strong foreground-background imbalance. Most pixels in an overhead image usually belong to background regions such as roads, fields, water, rooftops, or bare ground, while the target objects occupy only a small fraction of the image. This imbalance is especially problematic for dense one-stage detectors, where many easy background examples can dominate the training process. Focal Loss addresses this issue by down-weighting well-classified examples and focusing the training more strongly on hard examples, which makes it a suitable theoretical solution for foreground-background imbalance in dense object detection [6].

1.1.3 Oriented Object Detection

1.1.3.1 Limitations of Horizontal Bounding Boxes

Horizontal bounding boxes (HBBs) are the standard representation in general object detection, but they are often insufficient for aerial imagery. Since objects in overhead images can appear at arbitrary angles, an axis-aligned box may not match the actual object shape. For example, a rotated ship or vehicle can only be enclosed by a much larger horizontal rectangle, which includes a large amount of irrelevant background. This reduces the consistency between the visual features inside the box and the actual object, making classification and localization less reliable [1, 2].

This problem becomes more severe in dense scenes. In harbors, parking lots, airports, or traffic areas, multiple objects are often located close to each other. A single horizontal box may therefore cover not only the target object, but also neighboring instances. As a result, the extracted region features become ambiguous, because they may contain several objects at once. This also affects post-processing: when several horizontal boxes overlap strongly, non-maximum suppression (NMS) can incorrectly remove valid detections of nearby objects [1].

The geometric disadvantage of HBBs can also be illustrated quantitatively. If an oriented rectangle with width w, height h, and rotation angle θ is enclosed by its axis-aligned bounding box, the enclosing box has the approximate dimensions

$$W = w|\cos\theta| + h|\sin\theta|, \qquad H = w|\sin\theta| + h|\cos\theta|. \tag{1.1}$$

The ratio between the oriented object area and the enclosing horizontal box area is therefore

$$\mathrm{IoU}_{OBB,HBB} = \frac{wh}{WH}. \tag{1.2}$$

For an elongated object with an aspect ratio of $5:1$ rotated by 45°, this ratio drops to approximately 0.28. This means that more than two thirds of the horizontal box area consists of background. For even more elongated objects, such as ships, the mismatch becomes larger. Therefore, oriented bounding boxes (OBBs) are better suited for aerial imagery, because they can align with the object direction and describe the actual object extent more precisely [1, 2].

1.1.3.2 Oriented Bounding Box Parametrization

1.1.3.3 Angle Regression Challenges

1.1.3.4 Key Models

1.1.4 Datasets and Evaluation

1.1.4.1 Benchmark Datasets

1.1.4.2 Evaluation Metrics

1.2 Exercises and Homeworks

References

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